



**Spine** ➤ These help shade the stem and stop animals from eating the cactus. A spine is, in fact, a type of modified leaf with a small surface area to prevent water from evaporating.

*The ribs channel dew to the roots.*

### Monk's hood cactus

**Flakes** ➤ In a desert, the sun is very bright. White flakes grow on the stem of the cactus, helping to reflect light.

## Cacti shapes

### Columnar

Cacti, such as the saguaro, can grow 40 ft (12 m) tall, and have distinctive armlike side branches. Bats pollinate their flowers, which grow at the top of the plant.



### Clustering

The prickly pear cactus has flattened stems that grow in clusters. The red, prickly fruit must be peeled to remove the small spines.



### Globular

Many cacti, like the barrel cactus, have rounded stems. This maximizes water storage, while its vertical ridges direct every drop of water to the roots.



### Climbing

Some cacti live in forests, where they clamber up other plants for sunlight. The queen of the night cactus has huge flowers that bloom for just one night.



## Where do they grow?

### Deserts

Desert cacti have to survive extreme heat and light. Often large and spiny, they grow all over North and South America.



### Forests

Some cacti grow in shaded forests. Their stems do not need to store water, as the roots draw moisture from the air.



### Grasslands

Smaller cacti often grow in grasslands, where the grasses shade them in summer. Most are found in South America.

